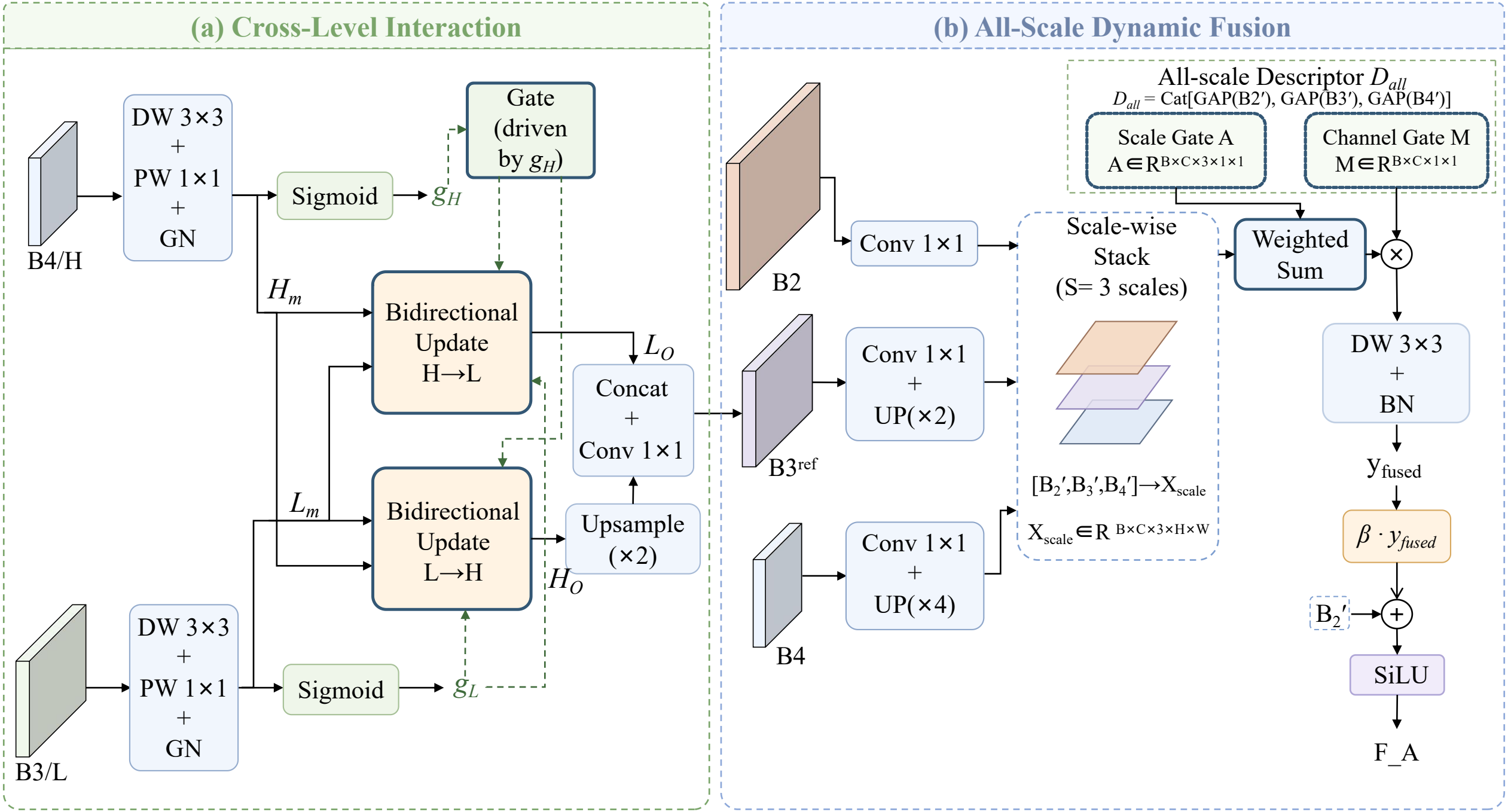

(a) Cross-Level Interaction
B4/H
DW 3×3 + PW 1×1 + GN
Sigmoid
g_H
Gate (driven by g_H)
H_m
Bidirectional Update H→L
L_O
Concat + Conv 1×1
L_m
Bidirectional Update L→H
Upsample (×2)
H_O
B3/L
DW 3×3 + PW 1×1 + GN
Sigmoid
g_L
(b) All-Scale Dynamic Fusion
All-scale Descriptor D_{all}
D_{all} = Cat[GAP(B2′), GAP(B3′), GAP(B4′)]
Scale Gate A
A∈R^{B×C×3×1×1}
Channel Gate M
M∈R^{B×C×1×1}
B2
Conv 1×1
B3^ref
Conv 1×1 + UP(×2)
B4
Conv 1×1 + UP(×4)
Scale-wise Stack (S= 3 scales)
[B2′,B3′,B4′]→X_scale
X_scale∈R^{B×C×3×H×W}
Weighted Sum
DW 3×3 + BN
y_fused
β · y_fused
B2′
SiLU
F_A